

Infinitesimal Strain in 3D

Straight forward generalisation of two dimensional theory.

— Initial location of a particle P is denoted by $P(x, y, z)$.

After deformation the new location is $P^*(x^*, y^*, z^*)$

Vector pointing from new position to old position is displacement vector $\vec{u}$.

$$\Rightarrow \vec{u}(P) = P - P^*$$

$$\text{or, } \langle u, v, w \rangle = \langle x, y, z \rangle - \langle x^*, y^*, z^* \rangle$$

} (35)

— Using the defn of strain components in 2D, the three normal strains are defined as

$$\epsilon_{xx} = \frac{\partial u}{\partial x}, \quad \epsilon_{yy} = \frac{\partial v}{\partial y} \quad \& \quad \epsilon_{zz} = \frac{\partial w}{\partial z} \quad \text{--- (36)}$$

eg:- ϵ_{xx} has the same interpretation as in 2D

i.e.; It is the fractional shortening of an infinitesimal line segment that is initially oriented along x-axis.

— The 3D strains are also defined in analogy with 2D case.

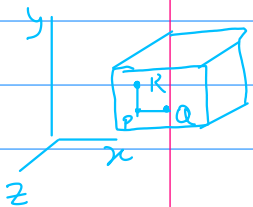
$$\left. \begin{aligned} \epsilon_{xy} &= \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right); \quad \epsilon_{xz} = \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right); \quad \epsilon_{yz} = \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \end{aligned} \right\} \text{--- (37)}$$

also, $\epsilon_{xy} = \epsilon_{yx}$; $\epsilon_{xz} = \epsilon_{zx}$ & $\epsilon_{yz} = \epsilon_{zy}$

These strains have the same interpretation in 2D.

eg:-

ϵ_{xy} represents the angle formed by two infinitesimally small line segments start at point (x, y, z) & lie along the x & y-directions respectively.



The proof of the fact that ϵ_{xy} is a correct interpretation of Eq(37) requires additional mathematical elaboration.

Because, the two lines will not necessarily continue to lie in xy-plane after deformation.

Let us start the 3D Generalisation of Eqns (1) (2) & (3) of 2D Case:

$$P^* = \vec{P} - \vec{u}(P) = (x, y, z) - [u(x, y, z), v(x, y, z), w(x, y, z)] \quad (38)$$

$$Q^* = \vec{Q} - \vec{u}(Q) = (x + \Delta x, y, z) - \left[\begin{array}{l} u(x + \Delta x, y, z) \\ + v(x + \Delta x, y, z) \\ + w(x + \Delta x, y, z) \end{array} \right] \quad (39)$$

$$R^* = \vec{R} - \vec{u}(R) = (x, y + \Delta y, z) - \left[\begin{array}{l} u(x, y + \Delta y, z) \\ v(x, y + \Delta y, z) \\ w(x, y + \Delta y, z) \end{array} \right] \quad (40)$$

Expanding the displacement components in a Taylor's Series about point (x, y, z) and Retaining terms that are linear increments of $\Delta x, \Delta y$. Yields

$$P^* = (x - u, y - v, z - w) \quad (41)$$

$$Q^* = \left(x + \Delta x - u - u \frac{\partial u}{\partial x} \Delta x, y - v - \frac{\partial v}{\partial x} \Delta x, z - w - \frac{\partial w}{\partial x} \Delta x \right) \quad (42)$$

$$R^* = \left(x - u - \frac{\partial u}{\partial y} \Delta y, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y, z - w - \frac{\partial w}{\partial y} \Delta y \right) \quad (43)$$

Here all the derivatives are evaluated at $P(x, y, z)$.

The angle formed by the Line Segment P^*Q^* and P^*R^* Can be Calculated as

$$(\vec{P^*Q^*}) \cdot (\vec{P^*R^*}) = |P^*Q^*| |P^*R^*| \cos \angle R^*P^*Q^* \quad (44)$$

From Eqn (41) & (42)

$$P^*Q^* = \left(\Delta x - \frac{\partial u}{\partial x} \Delta x, -\frac{\partial v}{\partial x} \Delta x, -\frac{\partial w}{\partial x} \Delta x \right) \quad (45)$$

$$\text{|| by } P^*R^* = \left(-\frac{\partial u}{\partial y} \Delta y, \Delta y - \frac{\partial v}{\partial y} \Delta y, -\frac{\partial w}{\partial y} \Delta y \right) \quad (46)$$

Neglecting the higher powers of displacement components.

$$\text{so, } (\vec{P^*Q^*}) \cdot (\vec{P^*R^*}) = \left(-\frac{\partial u}{\partial y} \Delta x \Delta y - \frac{\partial v}{\partial y} \Delta x \Delta y \right) \quad (47)$$

The lengths the two components are

$$\vec{PQ}^* \cdot \vec{PQ}^* = |\vec{PQ}^*|^2 = \left(\Delta x - \frac{\partial u}{\partial x} \Delta x \right)^2$$

Neglecting
higher powers
of gradient

$$\therefore |\vec{PQ}^*| = \Delta x \left(1 - \frac{\partial u}{\partial x} \right)$$

$$\text{Similarly } |\vec{PR}^*| = \Delta y \left(1 - \frac{\partial v}{\partial y} \right)$$

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$$\vec{PQ}^* \cdot \vec{PR}^* = -\Delta x \cdot \Delta y \left(1 - \frac{\partial u}{\partial x} \right) \frac{\partial u}{\partial y} - \Delta x \Delta y \left(1 - \frac{\partial v}{\partial y} \right) \frac{\partial v}{\partial x} + \frac{\partial u}{\partial x} \Delta x \frac{\partial v}{\partial y} \Delta y$$

Neglecting the products of higher than one order of disp grad

$$= -\Delta x \Delta y \frac{\partial u}{\partial y} - \Delta x \Delta y \frac{\partial v}{\partial x} = -\Delta x \Delta y \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)$$

$$\cos \angle R^* P^* Q^* = \frac{\vec{PQ}^* \cdot \vec{PR}^*}{|\vec{PQ}^*| |\vec{PR}^*|}$$

Now using Eqn (44)

$$\cos \angle R^* P^* Q^* = \frac{-\left[\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right] \Delta x \Delta y}{\Delta x \left(1 - \frac{\partial u}{\partial x} \right) \cdot \Delta y \left(1 - \frac{\partial v}{\partial y} \right)}$$

$$= \frac{-\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)}{\left(1 - \frac{\partial u}{\partial x} \right) \left(1 - \frac{\partial v}{\partial y} \right)}$$

$$= \frac{-\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)}{1 - \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)}$$

$\forall x \ll 1$

$$\therefore \frac{x}{1-x} = x(1-x)^{-1} = x(1+x) = x + x^2 \approx 0$$

$$= -\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) x \cdot \left(1 - \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)^{-1}$$

$$= -\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) \left[1 + \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) \right]$$

Biological Expansion.

$$= -\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) - \underbrace{\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)}$$

$$= -\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)$$

Higher Powers are neglected.

logic - food is indigestible

$\Rightarrow \cos \angle R^*P^*Q^*$ is close to 90° .

let the increase by δ

$$\text{Then } \angle R^*P^*Q^* = (90 + \delta)$$

$$\therefore \cos(90 + \delta) = -\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$$

$$\Rightarrow -\sin \delta = -\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$$

$$\therefore \delta = \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$$

$$\epsilon_{xy} = \frac{1}{2} L^+_{\Delta x \Delta y \rightarrow 0} (\angle R^* P^* Q^* - 90^\circ)$$

$$= \frac{1}{2} L^+_{\Delta x \Delta y \rightarrow 0} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)$$

$$\boxed{\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)}$$

This proof thus Verifies that Shear Strains are a measure of angular distortion in 3D.

The nine Components so defined in (36) & (37) Can be written in Matrix form as under .

$$\epsilon = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{xz} & \epsilon_{yz} & \epsilon_{zz} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) & \frac{\partial v}{\partial y} & \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) & \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) & \frac{\partial w}{\partial z} \end{bmatrix} \quad \text{--- (52)}$$

As Seen in 2D Case, here also the Strain matrix is equal to the Symmetric part of the displacement gradient ∇u

$$\therefore \epsilon = \text{Sym}(\nabla \vec{u}) = \frac{1}{2} \left[\nabla \vec{u} + (\nabla \vec{u})^T \right] \quad \text{--- (53)}$$

$$\text{Where } \nabla u = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix} \quad \text{--- (54)}$$

The antisymmetric part of the displacement gradient is again defined to be the rotation tensor, ω .
 whose components are given by

$$\omega = \frac{1}{2} [\nabla \vec{u} - (\nabla \vec{u})^T]$$

$$= \begin{bmatrix} \omega_{xx} & \omega_{xy} & \omega_{xz} \\ \omega_{yx} & \omega_{yy} & \omega_{yz} \\ \omega_{zx} & \omega_{zy} & \omega_{zz} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \frac{1}{2} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ -\frac{1}{2} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) & 0 & \frac{1}{2} \left(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \right) \\ -\frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) & -\frac{1}{2} \left(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \right) & 0 \end{bmatrix} \quad \text{--- (55)}$$

- The 3D demonstration that Strain is indeed a second order tensor, follows exactly the same lines as given in Eqn(22) Chain Rule.
- In particular, there will always be three mutually $\perp^r$ directions in which the normal strain takes on locally stationary values, typically a maximum, a minimum and a Saddle point. These three strains are denoted by $\epsilon_1 \geq \epsilon_2 \geq \epsilon_3$.
- In the principal Co-ordinate system in which the three principal directions serve as the co-ordinate axes, the Shear Strains are zero & the Strain tensor is purely diagonal
 (The question whether or not the principal directions of Strain coincide with the principal direction of Stress depends on the Stress Strain law of the rock).

- The values of three principal strains are found by solving an equation analogous to

$$e^3 - \mathbb{I}_1 e^2 - \mathbb{I}_2 e - \mathbb{I}_3 = 0 \quad \text{--- (57)}$$

Where, the strain Invariants are defined by

$$I_1 = \text{trace}(\epsilon) = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz} \quad \text{—————} (58)$$

$$I_2 = \frac{1}{2} \left[\text{trace}(\epsilon^2) - \{\text{trace}(\epsilon)\}^2 \right]$$

$$= E_{xy}^2 + E_{yz}^2 + E_{zx}^2 - E_{xy}E_{yz} - E_{yz}E_{zx} - E_{zx}E_{xy} \quad \text{--- (59)}$$

$$I_3 = \det(E)$$

$$= \epsilon_{xx}\epsilon_{yy}\epsilon_{zz} + 2\epsilon_{xy}\epsilon_{xz}\epsilon_{yz} - \epsilon_{xx}\epsilon_{yz}^2 - \epsilon_{yy}\epsilon_{xz}^2 - \epsilon_{zz}\epsilon_{xy}^2$$

- ① The first invariant of Strain has the physical interpretation of representing the Volumetric Strain in the Vicinity of (x, y, z) .

Proof:- Let a small cube of length L be aligned to the principal strain co-ordinate axes.

Volume Before deformation $V = L^3$.

After deformation the edges of the cube will be

$$L(1-\epsilon_1), \quad L(1-\epsilon_2), \quad L(1-\epsilon_3).$$

$$\therefore V^* = L^3 (1 - \epsilon_1)(1 - \epsilon_2)(1 - \epsilon_3).$$

$$\left. \begin{array}{l} \text{Volumetric} \\ \text{Strain} \end{array} \right\} = \left. \begin{array}{l} \text{Fractional decrease} \\ \text{in vol of cub.} \end{array} \right\} = \frac{V - V^*}{V}$$

$$= \frac{L^3 - L^3(1-\epsilon_1)(1-\epsilon_2)(1-\epsilon_3)}{1^3}$$

$$E_V R = \frac{1 - (1 - \epsilon_1 - \epsilon_2 - \epsilon_3 \dots)}{\epsilon_1 + \epsilon_2 + \epsilon_3} = \frac{1}{1} \quad \text{Neglecting Higher terms} \quad (61)$$

- In an approach similar to stresses, the strain tensor can be decomposed into isotropic & deviatoric part.

$$\epsilon = \epsilon^{iso} + \epsilon^{dev} \quad \text{--- (62)}$$

$$\epsilon^{iso} = \frac{1}{3} I_1 I \quad \leftarrow I = \text{Identity matrix.}$$

$$= \frac{1}{3} \epsilon_v I \quad \text{--- (63)}$$

$$\epsilon^{dev} = \epsilon - \epsilon^{iso}$$

$$= \epsilon - \frac{1}{3} \epsilon_v I \quad \text{--- (64)}$$

Deviatoric strain is sometime denoted by e

$$\therefore e = \epsilon - \frac{1}{3} \epsilon_v I = \begin{bmatrix} \epsilon_{xx} - \frac{1}{3} \epsilon_v & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} - \frac{1}{3} \epsilon_v & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} - \frac{1}{3} \epsilon_v \end{bmatrix} \quad \text{--- (65)}$$

The mean normal strain is

$$\text{defined by } \epsilon_m = \frac{1}{3} I_1 = \frac{1}{3} \epsilon_v.$$

- The principal axes of the deviatoric strain coincide with the principal axes of strain, and the magnitudes of the principal deviatoric strains are

$$e_1 = \epsilon_1 - \epsilon_m, \quad e_2 = \epsilon_2 - \epsilon_m, \quad e_3 = \epsilon_3 - \epsilon_m \quad \text{--- (67)}$$

Determination of Principal Stresses or Strains from measurements:

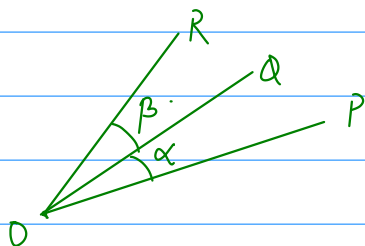
- measurement of Strain gives idea of Stress, provided the Stress Strain relation of rocks are known.
- Instruments
 - eg:- Strain gauges
 - Displacement meters } → Capable of Measuring Normal Strain in given direction.
- Solution to such problems is to find the normal strains then derive the principal strains.
- In 2D there are three independent Strain Components
↓
3-measurements will be sufficient to find those components.

Alternatively we can say:-

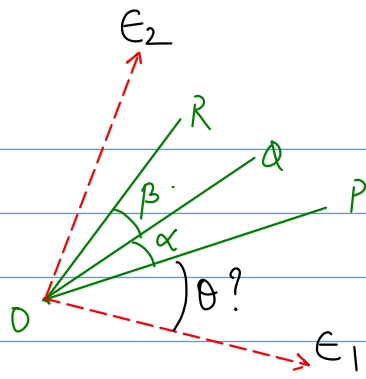
3-pieces of data are required to

- (i) estimate the two principal strains
- (ii) the angle of rotation of one principal direction wrt some arbitrary direction.

- Assume that we have measured the normal strain in 3-directions P, Q & R.



The angle of rotation between OP, OQ and OR is known.



However, the angle of rotation say θ between OP & the direction of major principal direction is not known.

if this is so, let it be θ .

$$\epsilon_{pp} = \frac{1}{2}(\epsilon_1 + \epsilon_2) + \frac{1}{2}(\epsilon_1 - \epsilon_2) \cos 2\theta$$

$$\epsilon_{qq} = \frac{1}{2}(\epsilon_1 + \epsilon_2) + \frac{1}{2}(\epsilon_1 - \epsilon_2) \cos 2(\theta + \alpha)$$

$$\epsilon_{rr} = \frac{1}{2}(\epsilon_1 + \epsilon_2) + \frac{1}{2}(\epsilon_1 - \epsilon_2) \cos 2(\theta + \alpha + \beta)$$

These 3-equations can be solved for θ .

Case-I:- For the commonly used configuration in which $\alpha = \beta = 45^\circ$, the solution is

$$\epsilon_1 + \epsilon_2 = \epsilon_{pp} + \epsilon_{qq}$$

$$\epsilon_1 - \epsilon_2 = \left[(\epsilon_{pp} - 2\epsilon_{qq} + \epsilon_{rr})^2 + (\epsilon_{pp} - \epsilon_{rr})^2 \right]^{1/2}$$

$$\theta = \frac{1}{2} \tan^{-1} \left[\frac{\epsilon_{pp} - 2\epsilon_{qq} + \epsilon_{rr}}{\epsilon_{pp} - \epsilon_{rr}} \right]$$

Case-II:- Another common case is $\alpha = \beta = 60^\circ$

$$\epsilon_1 + \epsilon_2 = \frac{2}{3}(\epsilon_{pp} + \epsilon_{qq} + \epsilon_{rr})$$

$$\epsilon_1 - \epsilon_2 = \left\{ \frac{4}{3}(\epsilon_{qq} - \epsilon_{rr})^2 + \frac{4}{9}(2\epsilon_{pp} - \epsilon_{qq} - \epsilon_{rr})^2 \right\}^{1/2}$$

$$\theta = \frac{1}{2} \tan^{-1} \left[\frac{\sqrt{3}(\epsilon_{qq} - \epsilon_{rr})}{\epsilon_{qq} + \epsilon_{rr} - 2\epsilon_{pp}} \right]$$